

VIVEK KOTHAPALLI

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PROFESSIONAL SUMMARY

MSc Artificial Intelligence with Business Strategy graduate from Aston University with hands-on experience in Python, machine learning, NLP and Generative AI. Built and deployed ML applications for bank fraud detection and customer churn prediction, and a Retrieval-Augmented Generation (RAG) banking assistant using OpenAI embeddings, LangChain and ChromaDB. Experienced in data preparation, model evaluation, semantic retrieval, LLM integration, Streamlit deployment, Git and GitHub. Seeking graduate AI/ML engineering opportunities.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy

Machine Learning: Scikit-learn, Logistic Regression, Random Forest, classification, model evaluation, ROC-AUC

Generative AI & NLP: RAG, LLMs, prompt engineering, OpenAI API, embeddings, LangChain, ChromaDB, NLP, Rasa

Deep Learning & Analysis: TensorFlow, neural networks, Deep Learning, data cleaning, preprocessing, EDA, Matplotlib

Deployment & Development: Streamlit, Git, GitHub

EXPERIENCE

Data Science Intern | Oeson | Jan 2025 - Apr 2025

- Worked on practical data science and machine learning tasks using Python, Pandas and NumPy.
- Performed data cleaning, preprocessing and exploratory data analysis (EDA), identifying patterns and preparing datasets for machine learning.
- Applied machine learning concepts to datasets and strengthened understanding of the end-to-end data science workflow.

PROJECTS

AI Banking Knowledge Assistant - RAG & Generative AI | Portfolio Project | 2026

- Built and deployed a Retrieval-Augmented Generation (RAG) application that answers banking questions using FCA source material with grounded LLM responses and source/page attribution.
- Built a document ingestion and retrieval pipeline using OpenAI text-embedding-3-small, LangChain, ChromaDB semantic retrieval (k=4) and relevance filtering.
- Evaluated retrieval/refusal behaviour on an initial 8-question test set, achieving 8/8 correct decisions across in-domain banking queries and out-of-domain questions.

Live Demo: vivek-rag-ai-assistant-project.streamlit.app | GitHub: [banking-rag-assistant](#)

Bank Fraud Detection - Machine Learning Application | Portfolio Project | 2026

- Built an end-to-end application to estimate transaction fraud risk from transaction and account-related features using synthetic transaction data.
- Compared Logistic Regression and Random Forest using precision, recall, F1-score and ROC-AUC; Logistic Regression achieved ROC-AUC 0.889 and recall 0.825.
- Deployed an interactive Streamlit application for real-time fraud-risk predictions with user-facing risk indicators.

Live Demo: vivek-bank-fraud-detection.streamlit.app | **GitHub:** [Repository](#)

Customer Churn Prediction - Machine Learning Application | Portfolio Project | 2026

- Built an end-to-end machine learning application to predict customer churn risk using customer behaviour and account-related features from synthetic data.
- Performed data preparation, model training and evaluation using Logistic Regression and Random Forest.
- Deployed an interactive Streamlit application that accepts customer information and returns churn probability.

Live Demo: vivek-customer-churn-prediction.streamlit.app | **GitHub:** [Repository](#)

NLP-Based Healthcare Appointment Chatbot | MSc Project, Aston University | Jun 2024 - Oct 2024

- Built a Python-based chatbot using Rasa to support healthcare appointment booking, rescheduling and cancellation.
- Used Rasa NLU for intent recognition and entity extraction, with custom Python logic for cases such as unavailable appointment slots.
- Integrated doctor availability data to support dynamic appointment interactions.

EDUCATION

MSc Artificial Intelligence with Business Strategy | Aston University, Birmingham, UK | Sep 2023 - Oct 2024

- Relevant study included Machine Learning, Deep Learning, Mathematics for AI and strategic applications of emerging technologies.
- MSc project focused on an NLP-based healthcare chatbot for appointment scheduling, rescheduling and cancellation.

Bachelor's Degree - Electronics and Communication Engineering | Aditya College of Engineering & Technology, India | 2015 - 2019

ADDITIONAL INFORMATION

Languages: English, Telugu